



NEWSLETTER

APRIL 2025



CELEBRATING ONE YEAR OF AUS RACING

MULTIDISCIPLINARY
STUDENTS PIONEERING
IN THE FIELD OF
RACING AND MAKING
VISIONS A REALITY

ITERATING THE CHASSIS

THE REAL REASON
BEHIND MAKING
MULTIPLE CHASSIS

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WHAT IS AUS RACING

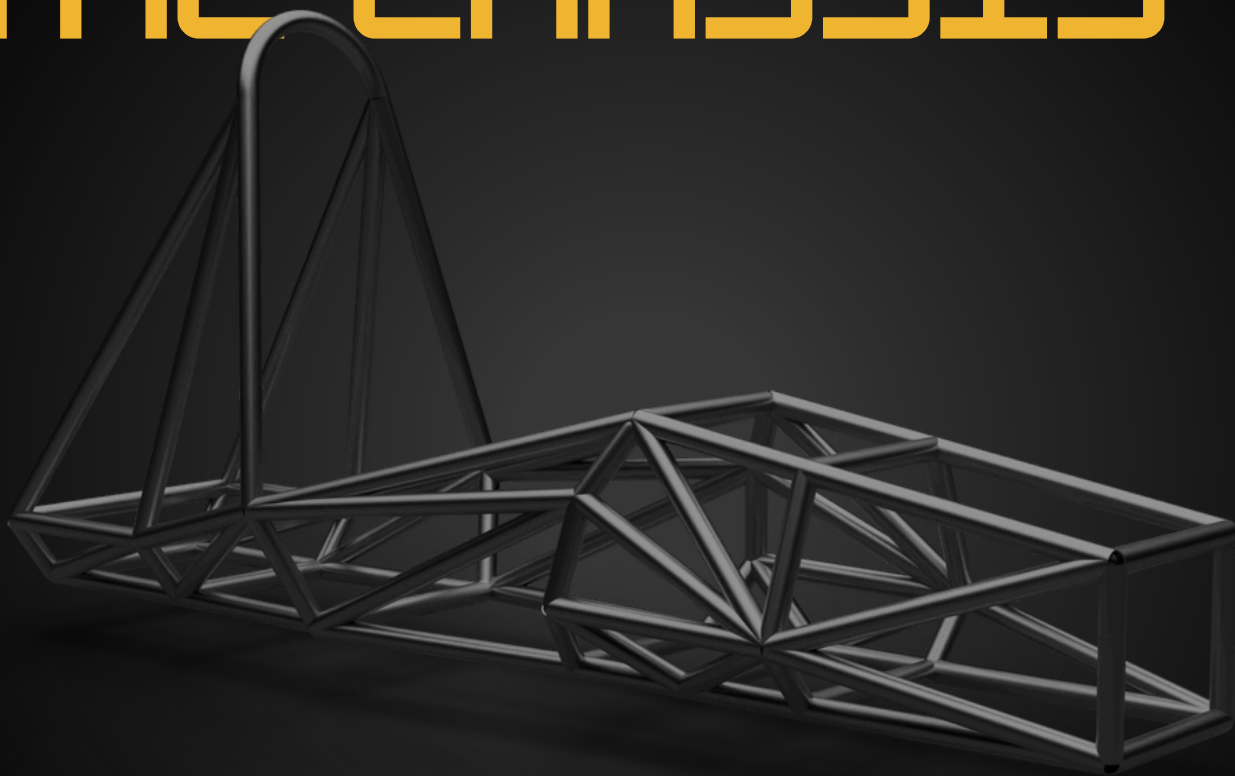
AUS Racing is the Formula Student team from the American University of Sharjah. Formula Student is an international engineering design competition that graciously provides a platform to aspiring engineers to build an open-wheel, single seater racecar and simulate all the different nuances that culminate to account for its structural integrity, performance, economic feasibility and industrial standards.

Students compete in different static and dynamic events to assess the car, thus testing the engineering design, management skills and the overall team spirit of the contesting teams. It is an esteemed competition that unlocks new horizons for AUSers

to venture into as we progress into a new era of technological competence and innovation.

Here at AUS Racing, we hope to open up opportunities for students for the foreseeable future to take part in a project that will teach them engineering, management and budgeting. In essence, real world engineering.

ITERATING THE CHASSIS



The chassis holds everything together. Any change to any part on the car directly affects the chassis.

Therefore, the chassis will go through numerous iterations before arriving at a final design that can be manufactured.

Not only should the chassis be able to hold everything together, it

also needs to have a high resistance to twisting.

Any deflection in the chassis will change the position of the suspension which has been carefully designed to optimize performance and handling.

Making a stiff chassis adds weight to the car. Therefore, it's imperative to find the right balance.

MAKING IT AERODYNAMIC



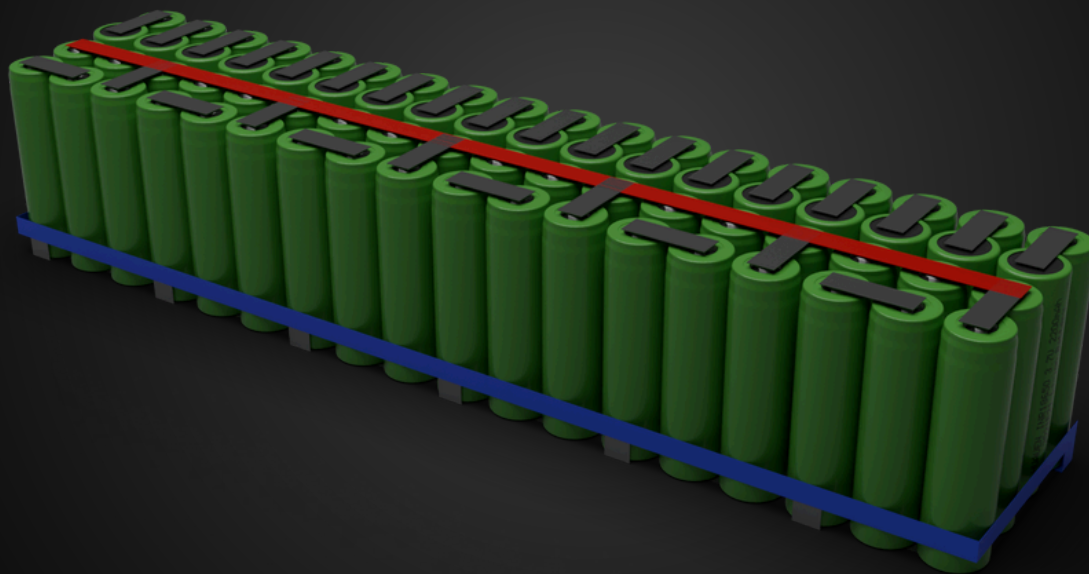
An aerodynamic car reduces drag, or friction due to air, which increases performance. In fact, reducing drag is one of the biggest challenges faced by the fastest cars in the world.

However, at what point does an extremely aerodynamic body, which reduces drag significantly, add too much weight to the car?

Although there is a fairly straightforward way to validate performance, by running simulations, it is extremely time taking.

Every iteration of the bodywork needs to be analyzed in Ansys Fluent. This data then needs to be fed into IPG Carmaker and simulated further.

TABS ON VOLTAGE



The voltage of every cell is constantly monitored and controlled. This is paramount for safety and increasing the longevity of the cells and improve the overall performance of the battery.

For instance, while charging the battery, it is crucial that each cell gets the same amount of energy. This is done by cell balancing.

Similarly, while discharging, or

while drawing power from the battery, energy from each cell needs to reduce at the same rate to ensure all the cells have the same stored energy at all times.

The voltage of the cells is also a clear indicator of how much energy is left in the cell as well as if there is anything going wrong.

An unexpected voltage drop in a cell indicates that there is probably something wrong.

ONE YEAR OF AUS RACING



A year ago, a team of ambitious students from the American University of Sharjah banded together to materialise their vision. And now, AUS Racing is officially registered for Formula Student UK, going strong with a team of 50 driven, multidisciplinary students.

From university workshops to sponsorships and meetings with top-notch companies, we have progressed impressively.

As a result of our passion and toil, we have had the privilege of attaining sponsorships by multinational corporations Ansys, IPG Automotives and Automech, while also setting a meeting with BMW AGMC.

We are striving to maintain the rigour, creativity and enthusiasm that has gotten us so far, and thus be pioneers for AUS students in the world of Formula Student.

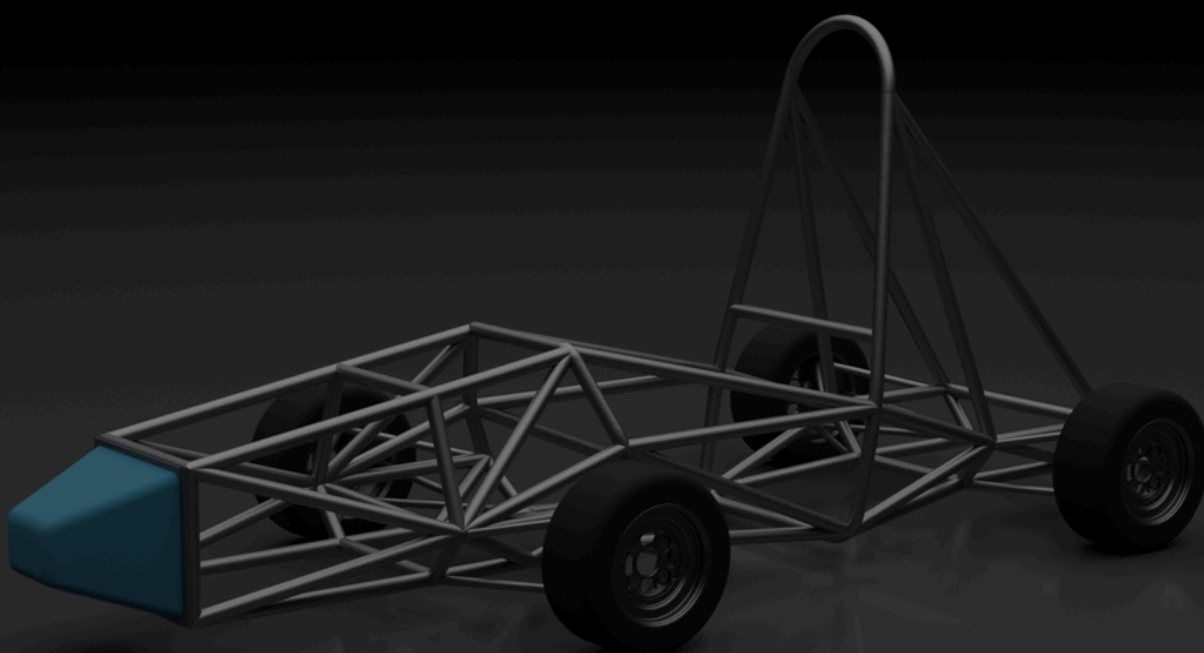
HOW YOU CAN SUPPORT US

What if you had the front row seat to engineering pioneers at work?

What if you were the reason these budding innovators were changing the future?

What if your institution could get their message to the rest of the world through Formula Student?

If your company's answer to any of these questions is yes, contact aaaa@aus.edu to learn more about getting involved with the AUS Racing team.



AUS

American
University
of Sharjah

GOLD

Ansys

BRONZE

AUTOMECH
GROUP

IPG